Chapter 3 Overview

	3.1 Linear Transformation
Learning Outcomes	• What is a **linear transformation**? • What is the standard matrix of a linear transformation? • What are the linearity properties of a linear transformation? • How to use basis to determine linear transformation? • How to use stacking method to find standard matrix of a linear transformation?
Key Points	• A matrix can be regarded as a mapping (linear transformation) on n-vector through multiplication. • A linear transformation can be given by formula or by its standard matrix. • The images of the standard basis vectors under a linear transformation are the columns of the standard matrix of the transformation. • A linear transformation preserves vector addition, scalar multiplication and linear combination. • A linear transformation is completely determined if the images of a basis for the domain are known.
	3.2 Eigenvalues and Eigenvectors
Learning Outcomes	• What are **eigenvalue** and **eigenvectors**? • How to find eigenvalues of a matrix? • What is the characteristic polynomial of a matrix? • How is eigenvalue related to singularity of a matrix?
Key Points	• If $Ax = \lambda x$ where x is a non-zero vector, then x is an eigenvector and λ is the associated eigenvalue of A. • A scalar multiple of an eigenvector is again an eigenvector associated with the same eigenvalue. • λ is an eigenvalue of A $\leftrightarrow$ $\det(\lambda I - A) = 0$. • The roots of the characteristic polynomial $\det(\lambda I - A)$ of A are the eigenvalues of A. • If 0 is an eigenvalue of A, then A is singular. Otherwise, A is non-singular.
	3.3 Eigenspaces
Learning Outcomes	• What is **eigenspace**? • How to find eigenvectors of a matrix? • How to find basis for eigenspace of a matrix? • What is the multiplicity of an eigenvalue? • How is the multiplicity related to the dimension of eigenspace?
Key Points	• Given an eigenvalue λ , the solution space of the homogeneous system $(\lambda I - A)x = 0$ is called the eigenspace of A associated to λ. • Any non-trivial solution of the above system is an eigenvector of A associated with λ. • The n x n identity matrix $\boldsymbol{I}$ has only one eigenvalue 1 with eigenspace = $\mathbf{R}^n$. • Dimension of the eigenspace of A associated to λ is less than or equal to the multiplicity of λ.

	3.4 Diagonalizable Matrices
Learning Outcomes	• What is a **diagonalizable** matrix? • How is a diagonalizable matrix related to eigenvalues and eigenvectors?
Key Points	• If A can be written as $A = PDP^{-1}$, where D is a diagonal matrix, then A is a diagonalizable matrix and P is said to diagonalizes A. • If an n x n matrix A has n linearly independent eigenvectors, then A is diagonalizable. • If we can only find fewer than n linearly independent eigenvectors for an n x n matrix A, then A is not diagonalizable. • $AB = A(b_1\ \ b_2\ \ \cdots\ \ b_n) = (Ab_1\ \ Ab_2\ \ \cdots\ \ Ab_n)$ where $b_1\ \ b_2\ \ \cdots\ \ b_n$ are the columns of B. • $BD = (b_1\ \ b_2\ \ \cdots\ \ b_n)D = (d_1b_1\ \ d_2b_2\ \ \cdots\ \ d_nb_n)$ where D is a diagonal matrix with diagonal entries d_1 , d_2 , $\cdots$, d_n.
	3.5 Diagonalization
Learning Outcomes	• How to determine if a matrix is diagonalizable? • How to diagonalize a matrix?
Key Points	• If the union of the basis from all the eigenspaces of an n x n matrix contains n eigenvectors, then the matrix is diagonalizable. • If an n x n matrix $A = PDP^{-1}$, then P is made up of the n eigenvectors as its columns, and the n diagonal entries of D are the eigenvalues. • The matrix P above is not unique, as there are many choices of eigenvectors. • If an n x n matrix A has n distinct eigenvalues, then A is diagonalizable. • It is possible for an n x n diagonalizable matrix to have fewer than n distinct eigenvalues. • If $\dim E_\lambda$ = multiplicity of λ for all eigenvalues of A, then A is diagonalizable. • If $\dim E_\lambda$ < multiplicity of λ for one (or more) of the eigenvalues of A, then A is not diagonalizable.
	3.6 Powers of Matrices
Learning Outcomes	• How to compute powers of matrix using diagonalization? • How to solve iterative system problems using diagonalization?
Key Points	• In an iterative system, we have $\boldsymbol{x_k} = \boldsymbol{Ax_{k-1}}$ where $\boldsymbol{x_k}$ is the stage k vector, and A is the iterative matrix. Then $\boldsymbol{x_n} = \boldsymbol{A}^n\boldsymbol{x_0}$ where $\boldsymbol{x_0}$ is the initial stage vector. • If A is a diagonalizable matrix with $\boldsymbol{P^{-1}AP = D}$, then $\boldsymbol{A^m = PD^mP^{-1}}$.